

Tube Radius Calculation Summary

Source

This note summarizes the double-integrator conformal tube radius used by `fine_tuning/scripts/dconformal_contraction_verify.py`. The numerical example below uses the current report snapshot:

`fine_tuning/plots/dconformal_contraction_verify_1849.json`.

For that run:

$$\begin{aligned} \text{sample_id} &= 1858, & \text{calibration samples} &= 1850, \\ s_u &= 0.836826, & s_x &= 0.268212, \\ q_u &= 0.651208, & q_x &= 0.386208, \\ \delta_u &= 0.15, & \delta_x &= 0.15. \end{aligned}$$

The reported joint probability lower bound is therefore

$$1 - \delta_u - \delta_x = 1 - 0.15 - 0.15 = 0.70.$$

Tube Formula

The code computes the radius at each time sample as

$$r(t) = \sqrt{\frac{\lambda_{\max}}{\lambda_{\min}}} \|x(0) - x_d(0)\| e^{-\alpha t} + \frac{\sqrt{\lambda_{\max}}}{\alpha \sqrt{\lambda_{\min}}} (\|B\|q_u + \|BK\|q_x) (1 - e^{-\alpha t}).$$

The first term is the decaying initial tracking error. The second term is the conformal uncertainty contribution from control mismatch and state/reference mismatch.

Fixed Values for the Current LQR Setup

These values are fixed as long as the double-integrator model and LQR weights in `fine_tuning/scripts/lqr.py` are unchanged:

$$\begin{aligned} \lambda_{\min}(P) &= 0.6126217787, \\ \lambda_{\max}(P) &= 9.8122021157, \\ \sqrt{\lambda_{\max}/\lambda_{\min}} &= 4.0020916201, \\ \alpha &= 1.5310952658, \\ \|B\|_2 &= 1.1, \\ \|BK\|_2 &= 3.9937691073. \end{aligned}$$

The common multiplier in the steady uncertainty term is

$$\frac{\sqrt{\lambda_{\max}}}{\alpha \sqrt{\lambda_{\min}}} = 2.6138749883.$$

Numerical Calculation

Using the current quantiles:

$$q_u = 0.651208, \quad q_x = 0.386208,$$

the uncertainty contribution inside the parentheses is

$$\begin{aligned} \|B\|_{q_u} &= 1.1(0.651208) = 0.7163288, \\ \|BK\|_{q_x} &= 3.9937691073(0.386208) = 1.5424256, \\ \|B\|_{q_u} + \|BK\|_{q_x} &= 2.2587544. \end{aligned}$$

The asymptotic, or steady-state, radius contribution is

$$r_\infty = 2.6138749883(2.2587544) = 5.9041016 \text{ m.}$$

For the plotted trajectory, the largest tube radius occurs at approximately

$$t = 2.5 \text{ s.}$$

At this time,

$$e^{-\alpha t} = e^{-1.5310952658(2.5)} = 0.0217588.$$

The initial tracking error for this run is zero:

$$\|x(0) - x_d(0)\| = 0.$$

Therefore the decaying initial-error term contributes zero, and the finite-time radius is

$$\begin{aligned} r(2.5) &= 0 + 5.9041016(1 - 0.0217588) \\ &= 5.7756356 \text{ m.} \end{aligned}$$

This matches the reported maximum tube radius:

$$\text{max_tube_radius_m} = 5.775636.$$

What Is Fixed vs. What Can Change

Quantity	Status
B , damping, Q , R	Fixed unless the LQR/model setup in <code>lqr.py</code> is changed.
P , K , α	Derived from solving the Riccati equation; fixed only while A , B , Q , and R are fixed.
$\lambda_{\min}$, $\lambda_{\max}$	Eigenvalues of P ; change if P changes.
$\ B\ $, $\ BK\ $	Fixed for a fixed B and K .
q_u , q_x	Change with calibration CSV, sample count, δ_u , δ_x , scoring method, and trajectory scheduling.
s_u , s_x	Per-sample scores; change from sample to sample.
$\ x(0) - x_d(0)\ $	Initial tracking error; changes with initial condition.
t and horizon	Change with trajectory duration and simulation timestep.

Why the Radius Can Be Larger Than Expected

1. The actual tracking error is not the tube radius. For this run,

$$\max_t \|x(t) - x_d(t)\| = 0.306533 \text{ m},$$

while the certified tube radius is

$$5.775636 \text{ m}.$$

The tube is a conservative bound, not the realized error.

2. The state quantile q_x is not only position error. In this code,

$$x = [p_x, p_y, v_x, v_y],$$

so velocity mismatch contributes to q_x .

3. The q_x term is strongly amplified:

$$\|BK\|q_x = 3.9937691073q_x.$$

Then the whole uncertainty term is multiplied again by

$$\frac{\sqrt{\lambda_{\max}}}{\alpha\sqrt{\lambda_{\min}}} = 2.6138749883.$$

4. The conformal tube assumes the calibrated uncertainty can persist over time. That is why the uncertainty term grows with

$$1 - e^{-\alpha t}$$

toward a steady radius.

5. Changing the calibration quantiles changes the tube directly. Larger q_u or q_x increases the radius linearly.